ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

# SYSTEMS ANALYSIS AND DESIGN

## PAPER 1 · PRACTICE A

**MARKS: 100** **TIME: 3 hours**

**NC(V) Information Technology and Computer Science, Level 4. Section A: objective questions (30) · Section B: development methodologies, UML modelling, normalisation and database design, testing and implementation, security and quality (70).**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

## INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Draw diagrams neatly with a pencil and ruler, using the correct symbols.
5. Write neatly and legibly.

## SECTION A

## QUESTION 1

**1.1** Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 In a UML use case diagram, a stick figure represents a/an …

A class.
B actor.
C object.
D database.

1.1.2 A table is in first normal form (1NF) when …

A it has no partial dependencies.
B it has no transitive dependencies.
C it has no primary key.
D it has no repeating groups and every field holds a single value.

1.1.3 Testing individual modules on their own is called …

A integration testing.
B unit testing.
C system testing.
D acceptance testing.

1.1.4 Testing done by the client to confirm the system meets the requirements before sign-off is …

A unit testing.
B white-box testing.
C acceptance testing.
D regression testing.

1.1.5 Which UML diagram shows the order in which objects send messages to each other over time?

A Sequence diagram
B Class diagram
C Use case diagram
D Deployment diagram

1.1.6 The methodology in which work is done in short "sprints" with a daily stand-up meeting is …

A Waterfall.
B Scrum.
C V-model.
D Spiral only.

1.1.7 A dependency in which a non-key field depends on another non-key field is a …

A partial dependency.
B functional key.
C composite key.
D transitive dependency.

1.1.8 Testing in which the tester does not look at the program code is …

A black-box testing.
B white-box testing.
C code review.
D desk checking.

1.1.9 A primary key made up of two or more fields is a …

A foreign key.
B surrogate key.
C composite key.
D secondary key.

1.1.10 Retesting the whole system after a change to make sure nothing else broke is …

A load testing.
B alpha testing.
C regression testing.
D usability testing.

(10)

**1.2** Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 The Waterfall model is suitable when requirements are clear and unlikely to change.

1.2.2 In a use case diagram, the system boundary is drawn as a rectangle.

1.2.3 A table in 3NF may still contain transitive dependencies.

1.2.4 Load (stress) testing checks how the system performs with many users.

1.2.5 Encryption makes stored data unreadable without the key.

1.2.6 Beta testing is done by the developers only, inside the company.

1.2.7 An audit trail records who changed data and when.

1.2.8 In a class diagram, inheritance is shown with a hollow triangle arrow pointing to the parent class.

1.2.9 Normalisation increases data redundancy.

1.2.10 A disaster recovery plan explains how to restore the system after a major failure.

(10)

**1.3** Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

| | **COLUMN A** | | **COLUMN B** |
|---|---|---|---|
| 1.3.1 | Short description of a requirement from the user's point of view: "As a … I want … so that …" | A | non-functional requirement |
| 1.3.2 | Requirement that describes how well the system must perform, e.g. speed or security | B | integration testing |
| 1.3.3 | Requirement that describes what the system must do | C | authentication |
| 1.3.4 | Combining tested modules and testing them together | D | off-site backup |
| 1.3.5 | Proving the identity of a user, e.g. with a password or fingerprint | E | authorisation |
| 1.3.6 | Deciding what an authenticated user is allowed to do | F | functional requirement |
| 1.3.7 | Using two different methods to confirm identity, e.g. a password and an SMS code | G | service level agreement |
| 1.3.8 | Prioritised list of features still to be developed in Scrum | H | two-factor authentication |
| 1.3.9 | Agreement that sets the level of service a supplier must provide, e.g. 99% uptime | I | user story |
| 1.3.10 | Copy of data stored at another place to recover after a disaster | J | product backlog |

(10)

**TOTAL SECTION A: [30]**

## SECTION B

**Case study: QuickFix Appliance Repairs**
QuickFix repairs fridges, stoves and washing machines in Gqeberha. Customers phone to book a repair. A receptionist captures the customer's details and the appliance. The manager assigns a technician to each job. Technicians visit the customer, record the parts used and the hours worked, and the receptionist then sends an invoice. QuickFix wants a web-based system where customers can book online and track their repair, technicians can update jobs on their phones, and the manager can see reports on jobs and income.

## QUESTION 2: METHODOLOGIES AND REQUIREMENTS

**2.1** Compare the Waterfall and Agile methodologies under the headings: approach, client involvement and handling of changes. Recommend one for QuickFix with a reason. (8)

**2.2** List THREE functional and TWO non-functional requirements for the QuickFix system. (5)

**2.3** Write ONE user story for a technician. (2)

**[15]**

## QUESTION 3: UML MODELLING

**3.1** Draw a use case diagram for the QuickFix system with at least THREE actors and SIX use cases. Include the system boundary. (10)

**3.2** Draw a class diagram showing the classes Customer, Job and Technician with TWO attributes and ONE method each, and the multiplicity of the relationships. (6)

**3.3** Explain the purpose of an activity diagram. (2)

**[18]**

## QUESTION 4: NORMALISATION AND DATABASE DESIGN

**4.1** The receptionist currently keeps the following unnormalised data for jobs:

| JobNo | Date | CustNo | CustName | CustPhone | TechID | TechName | PartNo | PartDesc | Qty | Price |
|---|---|---|---|---|---|---|---|---|---|---|
| J101 | 03/05 | C12 | N. Mokoena | 0821234567 | T2 | S. Pillay | P7, P9 | Thermostat, Seal | 1, 2 | 450, 120 |
| J102 | 04/05 | C15 | J. Botha | 0739876543 | T1 | L. Dlamini | P3 | Element | 1 | 380 |
| J103 | 04/05 | C12 | N. Mokoena | 0821234567 | T2 | S. Pillay | P9 | Seal | 1 | 120 |

4.1.1 Identify TWO problems (anomalies) that this design could cause. (2)

4.1.2 Normalise the data to third normal form (3NF). Show the tables with their fields, and underline the primary keys and mark the foreign keys with an asterisk (*). (10)

4.1.3 Calculate the total value of the parts used on job J101. (2)

**4.2** Explain the difference between a partial dependency and a transitive dependency, using an example from the QuickFix data. (4)

**[18]**

## QUESTION 5: TESTING, IMPLEMENTATION, SECURITY AND QUALITY

**5.1** Describe the FOUR levels of testing the QuickFix system should go through, in order. (8)

**5.2** Give THREE security measures for a web-based system that stores customers' personal information, and link ONE of them to the requirements of POPIA. (4)

**5.3** Explain TWO things the analyst must plan for when the new system is implemented, other than the conversion method. (4)

**5.4** Name THREE characteristics of a high-quality information system. (3)

**[19]**

**TOTAL SECTION B: [70]**

**TOTAL: 100**

**END OF QUESTION PAPER**